AI CAREERS FOR
WOMEN
LINKEDIN FELLOWSHIP

PERSONAL FINANCIAL PLANNING FORECASTING MODEL

Project Domain

Excel /Power BI with Copilot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

This report is based on 20,000 individual financial records covering monthly income, spending categories, savings, debt, insurance, and emergency fund data. All benchmarks are sourced from SEBI (Securities and Exchange Board of India), RBI (Reserve Bank of India), FPSB India, and NPS guidelines — making this analysis reliable and India-specific.

The study identifies five critical problem areas across five age groups (18–64). For each problem, real-life scenarios, data evidence, and four actionable solutions are provided — all benchmarked against SEBI/RBI/FPSB India standards.

Key findings include: (1) Retirement savings gap in the 46–55 age group, with actual savings rate of 25.5% against the required 30%; (2) Pre-retirement debt and housing burden for the 56–64 group, with housing-to-income at 21.9% vs. the ≤15% benchmark; (3) Insufficient emergency funds during family formation for the 26–35 group; (4) Missing retirement investment habits in the prime compounding years for the 18–25 group; and (5) Systemic underinsurance across all age groups, with a monthly gap of ₹833 to ₹3,760.

Solutions proposed are grounded in NPS, ELSS, SCSS, PMVVY, and IRDAI guidelines. The consolidated action plan provides age-specific, prioritised steps for immediate (now), short-term (3 months), and medium-term (6 months) financial improvement.

This report is intended for educational and analytical purposes. Readers are advised to consult a SEBI-registered Investment Advisor (RIA) for personalised financial planning.

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CHAPTER 1

Introduction

This chapter presents real-world problem statements identified from a dataset of 20,000 individual financial records spanning five age groups (18–64 years). Each problem is accompanied by a real-life situational narrative, data evidence benchmarked against SEBI/RBI/FPSB India guidelines, and four structured solutions. The chapter also outlines the motivation for this study, the objectives pursued, and the scope of analysis.

1.1 Problem Statement

Personal financial planning in India faces five structural challenges identified from the dataset:

- Retirement Savings Gap (Age 46–55): Savings rate of 25.5% against the 30% SEBI/FPSB benchmark, with emergency funds at only 4.4 months against the required 12 months for pre-retirees. Housing-to-income at 22% exceeds the 20% benchmark, indicating ongoing home loan obligations that erode retirement corpus building.
- Pre-Retirement Debt & Housing Burden (Age 56–64): Housing-to-income ratio at 21.9% far exceeds the ≤15% benchmark for this life stage. Debt-to-income at 5.0% sits at the upper limit, while insurance spend of ₹1,440/month is structurally insufficient for rising post-60 medical costs.
- Insufficient Emergency Fund During Family Formation (Age 26–35): Despite a strong savings rate of 25.7%, emergency funds cover only 4.4 months against the 6-month minimum for households with dependents and EMIs. Average loan repayments of ₹2,012/month increase vulnerability to high-interest emergency debt.
- Missing Retirement Investment Habit in Prime Compounding Years (Age 18–25): A savings rate of 25.4% is commendable, yet no structured retirement instruments (NPS, ELSS) are evident. Savings in liquid accounts lose real value after inflation of 5–6% p.a., wasting the most powerful financial force — compounding over a 35–40 year horizon.
- Systemic Underinsurance Across All Age Groups (18–64): Actual insurance spend ranges from ₹1,438 to ₹1,484/month across all groups — significantly below the ₹2,300 to ₹5,200/month required for adequate term + health coverage. This creates a single-point-of-failure risk in every household’s financial plan.

1.2 Motivation

1. Real-World Relevance: This study is motivated by the gap between earnings and effective financial planning observed across Indian households. Rising living costs, unstructured savings, and inadequate insurance coverage affect financial stability across all income levels and age groups.

2. Data-Driven Evidence: The availability of 20,000 financial records with granular income, expense, savings, and insurance data enables precise identification of structural gaps — replacing anecdotal financial advice with benchmark-backed, evidence-based recommendations.

3. India-Specific Relevance: All benchmarks and solutions are anchored to SEBI, RBI, FPSB India, IRDAI, NPS Trust/PFRDA, and NHB guidelines — ensuring recommendations are legally and financially actionable within the Indian regulatory framework.

4. Lifecycle Applicability: The five age groups (18–25, 26–35, 36–45, 46–55, 56–64) cover the complete earning lifecycle, making the findings relevant to individuals at every stage of financial planning.

5. Alignment with Financial Literacy Goals: The study supports national financial literacy objectives by translating complex regulatory benchmarks into clear, actionable steps accessible to non-specialist readers.

1.3 Objectives

- Identify and document five structural financial problems from 20,000 individual financial records, each with real-life narrative context.
- Benchmark findings against SEBI, RBI, FPSB India, IRDAI, and NPS/PFRDA regulatory standards to ensure accuracy and credibility.
- Provide four evidence-based solutions per problem, covering savings instruments (NPS, ELSS, PPF, SCSS, PMVVY), insurance strategies (term, health floater, super top-up), and debt management approaches.
- Develop a consolidated age-group action plan with time-prioritised steps (immediate, 3 months, 6 months) for each of the five age segments.
- Present data evidence for each problem using actual dataset metrics versus regulatory benchmarks in a clear, comparable format.
- Enable individuals, financial advisors, and fintech platforms to use this analysis as a reference model for real-world financial planning decisions.

1.4 Scope of the Project

- This study analyses 20,000 individual financial records covering five age groups: 18–25, 26–35, 36–45, 46–55, and 56–64 years. Variables include monthly income, housing expense, savings rate, debt-to-income ratio, insurance spend, and emergency fund coverage in months.

- All benchmarks are sourced exclusively from Indian regulatory and advisory bodies: SEBI, RBI, FPSB India, IRDAI, NPS Trust/PFRDA, and National Housing Bank (NHB). No international benchmarks are applied.

- Five problem areas are analysed in depth: retirement savings gap, pre-retirement debt and housing burden, emergency fund insufficiency, missing retirement investment habits, and systemic underinsurance. Each problem includes background, data evidence, and four solutions.

- The scope does not include real-time market data, tax computation, personalised portfolio construction, or live financial advisory. This is an educational and analytical report, not a financial advisory product. Readers are advised to consult a SEBI-registered Investment Advisor (RIA) for personalised guidance.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Review of Existing Work

Personal financial management has been widely studied in the fields of finance and data analytics. Traditional approaches focus on budgeting techniques, ratio analysis, and financial planning models, where individuals track income, expenses, and savings to maintain financial stability.

In recent years, digital tools such as Excel-based financial trackers and Power BI dashboards have been used to visualize financial data and monitor spending patterns. Additionally, fintech applications like Mint and YNAB provide automated expense tracking and budgeting features.

Research in data analytics has also introduced predictive and AI-assisted financial models, which aim to forecast future financial conditions and improve decision-making

Existing Models, Techniques, and Tools

Several techniques and tools are relevant to this project:

- **Excel-based Models:**
  Used for data cleaning, ratio calculations, and financial analysis using formulas and pivot tables.

- **Power BI Dashboards:**
  Used for interactive visualization, KPI tracking, and financial reporting.

- **Financial Ratio Analysis:**
  Commonly used in finance to evaluate savings behaviour , debt levels, and investment performance.

- **Forecasting Techniques:**
  Basic forecasting using growth rates and trend analysis is widely used in financial planning.

- **AI Prompt Engineering & Chatbots:**
  AI tools such as Microsoft Copilot and are increasingly used for data interpretation, insight generation, and report automation.

2.2 Tools and Technologies Review

• Microsoft Excel — widely used for data cleaning, preprocessing, and financial analysis. In this project, Excel is used to calculate financial ratios, normalize data, and compute the Financial Health Score (FHS) using formulas, along with and conditional formatting for basic analysis.

• Microsoft Power BI — used for creating interactive dashboards and visualizations. It helps present key insights such as Financial Health Score distribution, risk categories, and future projections in a clear and user-friendly manner

• Microsoft Copilot (AI) — an AI assistant integrated with Microsoft tools, used to automate data analysis, generate insights, assist in formula creation, and improve report writing through natural language prompts.

• AI Chatbots (Google Gemini) — generative AI tools used for content creation, explanation of financial insights, report generation, and building a recommendation layer by converting numerical outputs into meaningful financial advice.

• Financial Modelling Techniques — includes ratio analysis, normalization, weighted scoring methods, and basic forecasting techniques used to evaluate and predict financial health. These techniques form the core logic of the model.

• Data Visualization Techniques — used to represent financial insights through charts, KPIs, and dashboards, making complex data easy to understand and interpret for users.

2.3 Gaps Identified

• Most existing financial models focus on **complex machine learning techniques**, while simpler, Excel-based and user-friendly solutions for non-technical users are underexplored.

• Existing personal finance tools are largely **descriptive in nature**, focusing on past transactions and current balances, with limited capability for **future forecasting and scenario-based analysis**.

• There is a lack of **integrated financial scoring systems** that combine multiple dimensions such as savings, debt, emergency funds, and investments into a single, easy-to-understand metric.

• Most solutions do not provide **personalized, AI-driven recommendations**, making it difficult for users to take actionable steps to improve their financial condition.

. Prompt engineering practices for financial and business use cases are still evolving and lack **standardized frameworks**, especially for generating consistent and reliable insights.

. There is limited integration of **macroeconomic factors such as inflation and interest rates** with personal financial data in existing models, reducing real-world applicability.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

Data Source / Input

The project uses a synthetic personal finance dataset of 1,000 individuals in Excel format, containing fields such as income, expenses, savings, EMI (loan burden), investments, emergency funds, and income growth rate.

It also incorporates derived variables (like savings ratio and debt ratio) and basic macroeconomic inputs such as inflation and interest rates to improve realism.

Data Cleaning

The dataset is preprocess using Excel techniques:

- Removal of duplicate records
- Handling missing or inconsistent values
- Standardizing data formats (monthly/annual values)
- Converting raw values into meaningful financial ratios
- Normalizing variables to a common scale (0–1) for fair comparison

Analysis / Processing

The core analysis is performed using a combination of Excel and AI tools:

- Financial Ratio Analysis: Calculation of savings ratio, debt ratio, emergency fund coverage, and investment ratio
- Weighted Scoring Model: Computation of Financial Health Score (FHS) based on predefined weights
- Forecasting: Estimation of future income, expenses, and savings using growth rates and inflation assumptions
- AI-Assisted Analysis: Use of tools like Microsoft Copilot and ChatGPT to generate insights, explain results, and provide recommendations

3.2 Modules / Components Used

Module 1 — Data Ingestion

Loading the personal finance dataset (1000 individuals) into Excel, including variables such as income, expenses, savings, EMI, investments, emergency funds, and external factors like inflation and interest rates.

Module 2 — Data Cleaning & Preprocessing

. Removing duplicates and handling missing values

- Standardizing formats (monthly/annual values)
- Creating derived variables such as savings ratio, debt ratio, and investment ratio
- Normalizing all variables to a common scale (0–1)

Module 3 — Financial Analysis & FHS Calculation

Applying financial formulas and weighted scoring to compute the **Financial Health Score (FHS)**:

- Ratio analysis (savings, debt, emergency fund, investment)
- Weighted model to generate a score from 0 to 100
- Classification into risk categories (Healthy, Moderate, Vulnerable, Risky)

Module 4 — Forecasting & Scenario Analysis

Estimating future financial condition by:

- Projecting income, expenses, and savings using growth rate and inflation
- Performing scenario analysis (e.g., income change, expense variation, EMI impact)
- Evaluating potential financial outcomes over a 5-year period

Module 5 — Visualization & Report Generation

- Microsoft Power BI dashboards for score distribution and trends
- Excel summary sheets with key metrics and forecasts

- Charts, KPIs, and reports for clear communication of insights

Module 6 — AI Integration

Using AI tools like Microsoft Copilot and ChatGPT for:

- Insight generation and explanation of results
- Automated report writing and summaries
- Personalized financial recommendations based on FHS

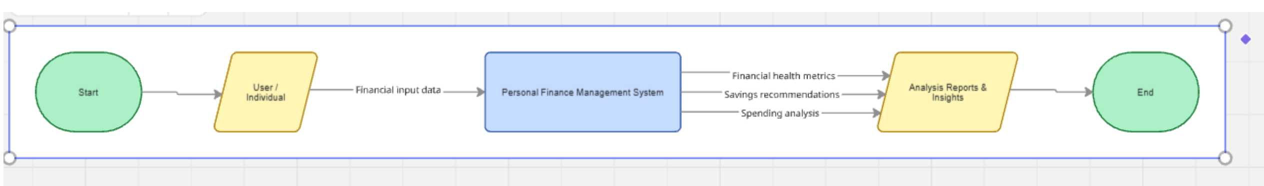
Module 7 — Decision Support System (Additional Module)

- Interpreting scores and risk levels
- Suggesting improvements (reduce expenses, increase savings, manage debt)
- Helping users make informed financial decisions

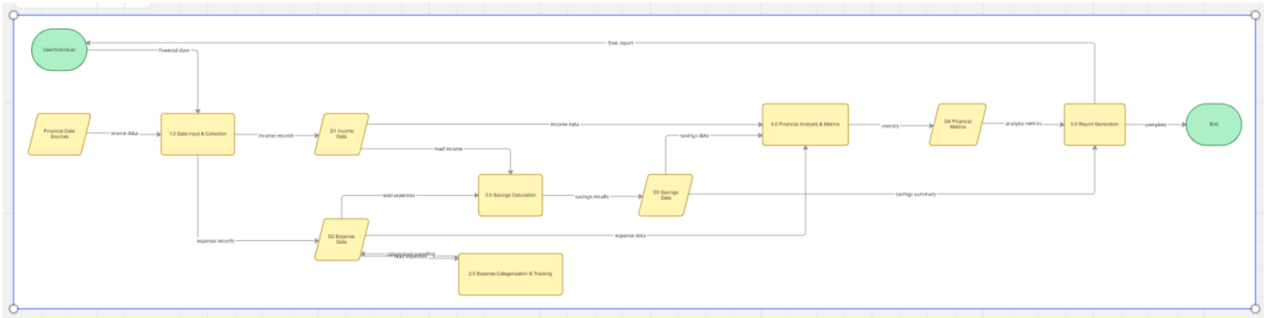
3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram



DFD LEVEL 1



3.4 Advantages

Automates repetitive data cleaning tasks
Saves significant manual effort by using preprocessing engines and AI-driven validation.

- Provides clear, visual insights
  Interactive Power BI dashboards make results easy to understand for non-technical stakeholders.

- Leverages AI tools (Copilot / Chatbots)
  Accelerates analysis, generates narrative summaries, simulates scenarios, and reduces human error.

- Scalable methodology
  Can be applied to larger datasets or extended to other domains such as SME finance or corporate risk forecasting.

- Cost-effective solution
  Built on widely available Microsoft 365 tools (Excel, Power BI, Copilot), minimizing additional software costs.

  - Personalized recommendations
  AI generates tailored action cards and improvement strategies for individuals, advisors, and fintech platforms.

- Transparent and auditable framework
  The Financial Health Score (FHS) formula is fully documented, interpretable, and reproducible, ensuring credibility.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	HP Victus with Intel Core i5
RAM	minimum 16 GB RAM (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 11 home
Microsoft Excel	Microsoft 365
Power BI Desktop	latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription, runs smoothly on Windows 11.
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Uncleaned data set

Age	Income (₹)	Rent (₹)	Loan Repayment (₹)	Insurance (₹)	Groceries (₹)	Transport (₹)	Eating Out (₹)	Entertainment (₹)	Utilities (₹)	Healthcare (₹)	Education (₹)	Miscellaneous (₹)	Desired Savings %	Desired Savings (₹)	Disposable Income (₹)	Pot. Savings Groceries (₹)	Pot. Savings Transport (₹)	Pot. Savings Eating Out (₹)	Pot. Savings Entertainment (₹)	Pot. Savings Utilities (₹)	Pot. Savings Healthcare (₹)	Pot. Savings Education (₹)	Pot. Savings Misc. (₹)	Total_Spending	Actual_Savings
49	44,637.25	13,391.17	0.00	2,206.49	6,658.77	2,636.97	1,651.80	1,536.18	2,911.79	1,546.91	0.00	831.53	13.89%	6,200.54	11,265.63	1,685.70	328.90	465.77	195.15	678.29	67.68	0.00	85.74	33,371.62	11,265.63
34	26,858.60	5,371.72	0.00	869.52	2,818.44	1,543.02	649.38	1,050.24	1,626.14	1,137.35	1,551.72	564.24	7.16%	1,923.18	9,676.82	540.31	119.35	141.87	234.13	286.67	6.60	56.31	97.39	17,181.78	9,676.82
35	50,367.61	7,555.14	4,612.10	2,201.80	6,313.22	3,221.40	1,513.81	1,723.31	3,368.46	2,178.52	3,160.03	628.37	14.00%	7,050.36	13,891.45	1,466.07	473.55	410.86	459.97	488.38	7.29	106.65	138.54	36,476.15	13,891.45
21	1,01,455.60	15,218.34	6,809.44	4,889.42	14,690.15	7,106.13	5,040.25	2,858.19	6,128.55	4,571.12	0.00	2,526.06	16.46%	16,694.97	31,617.95	1,875.93	762.02	1,241.02	320.19	1,389.82	193.50	0.00	296.04	69,837.65	31,617.95
52	24,875.28	4,975.06	3,112.61	635.91	3,034.33	1,276.16	692.83	660.19	1,092.69	1,169.10	1,445.22	515.51	7.53%	1,874.10	6,265.70	788.95	68.16	61.71	187.17	194.12	47.29	67.39	96.56	18,609.58	6,265.70
57	24,875.61	4,975.12	0.00	1,038.23	3,250.07	1,760.16	1,049.07	751.02	1,024.31	1,022.30	2,003.85	402.22	5.94%	1,478.67	7,599.27	957.62	521.68	135.93	148.95	86.57	13.14	33.96	96.68	17,276.34	7,599.27
53	1,06,119.69	21,223.94	0.00	4,360.20	12,790.39	6,345.74	4,390.91	2,626.77	6,202.55	3,807.10	0.00	2,776.77	17.34%	18,399.56	41,595.33	2,799.23	912.96	438.01	541.70	838.30	111.86	0.00	409.45	64,524.37	41,595.33
40	55,431.34	11,086.27	8,307.70	1,755.44	8,194.48	3,353.00	1,741.91	1,219.68	2,361.98	2,574.45	5,520.44	850.04	11.97%	6,636.41	8,465.94	1,059.50	838.16	466.18	327.90	353.41	92.74	116.12	71.52	46,965.40	8,465.94
32	20,606.73	6,182.02	3,562.49	1,018.59	3,066.87	1,170.79	688.06	418.23	921.23	830.24	1,939.23	446.07	6.30%	362.92	362.92	822.97	213.00	85.49	97.81	252.34	9.98	59.96	80.75	20,243.82	362.92
30	46,304.79	9,260.96	0.00	2,130.25	5,065.63	2,500.05	2,040.31	1,016.61	2,945.43	1,649.76	3,085.99	919.53	12.13%	5,615.10	15,690.26	962.02	380.27	598.88	188.31	793.64	34.87	39.74	203.15	30,614.53	15,690.26
31	20,706.82	3,106.02	0.00	1,031.78	3,037.27	1,459.28	451.41	785.29	1,548.08	819.10	0.00	410.03	8.35%	1,729.70	8,058.57	503.56	392.32	35.43	230.33	262.89	38.34	0.00	77.80	12,648.25	8,058.57
31	20,668.56	4,133.71	0.00	754.45	2,597.40	1,123.04	878.88	643.15	1,408.90	1,019.25	1,506.36	483.81	8.25%	1,704.19	6,119.61	554.20	267.30	119.57	123.96	271.59	1.50	29.70	94.49	14,548.95	6,119.61
54	36,407.22	10,922.17	4,583.25	970.79	4,685.67	2,503.09	1,328.96	822.88	2,335.94	1,153.75	1,940.13	570.12	7.53%	2,743.08	4,590.47	445.70	190.11	346.31	85.13	474.01	40.97	85.03	40.63	31,816.76	4,590.47
22	6,492.02	973.80	0.00	184.24	955.50	494.82	309.24	229.83	471.87	222.65	590.05	179.09	7.15%	464.29	1,880.93	180.47	38.25	59.83	49.16	60.74	3.92	20.42	46.56	4,611.09	1,880.93
39	7,547.85	1,509.57	0.00	330.19	1,032.73	568.24	266.44	277.47	510.60	235.71	521.98	133.49	5.46%	412.21	2,161.46	208.11	114.59	72.17	77.92	97.97	2.98	4.88	36.84	5,386.40	2,161.46
45	19,132.10	3,826.42	2,525.41	590.19	2,515.72	1,164.07	661.75	660.75	1,159.58	919.43	1,442.91	461.51	8.02%	1,533.96	3,204.35	323.14	108.91	183.12	101.13	314.59	17.27	66.99	92.37	15,927.74	3,204.35
62	13,342.21	2,668.44	2,540.48	352.89	1,920.10	961.38	622.27	331.70	985.63	531.56	1,043.20	286.26	8.14%	1,086.13	1,098.29	568.17	205.18	154.84	96.40	263.23	20.00	5.42	61.40	12,243.92	1,098.29
21	38,574.65	7,714.93	0.00	1,838.34	5,126.82	2,614.48	1,267.55	1,802.74	2,081.39	1,196.04	3,475.98	843.80	6.91%	2,666.09	10,612.58	862.85	202.78	217.66	113.22	435.10	6.84	41.43	179.80	27,962.07	10,612.58
55	14,509.13	4,352.74	829.44	438.60	1,506.25	811.70	515.58	414.23	766.34	656.65	1,449.96	285.02	7.24%	1,049.97	2,482.61	384.18	194.04	34.82	99.84	55.81	7.78	41.56	84.83	12,026.52	2,482.61
46	9,692.52	1,938.50	0.00	217.59	1,306.54	562.10	396.06	427.91	723.64	305.65	0.00	155.46	5.22%	505.94	3,659.07	138.17	102.74	118.04	69.69	66.64	3.07	0.00	19.68	6,033.45	3,659.07
54	96,903.57	19,380.71	9,481.17	3,585.07	14,257.03	6,271.72	2,829.05	3,879.58	4,066.69	3,956.93	6,227.47	2,131.73	13.55%	13,125.69	20,836.43	4,013.47	1,649.69	584.64	200.13	451.54	191.91	295.42	404.04	76,067.14	20,836.43
52	25,042.55	5,008.51	2,898.04	914.22	3,170.69	1,947.50	1,098.41	839.59	1,321.02	1,066.04	1,470.45	407.47	5.75%	1,439.44	4,900.62	859.65	335.61	218.41	161.42	355.09	50.18	33.31	70.79	20,141.93	4,900.62
37	31,665.25	9,499.58	0.00	1,034.49	4,019.47	2,180.72	1,373.34	876.08	1,686.60	1,579.08	0.00	623.56	6.84%	2,164.46	8,792.34	1,182.46	273.20	126.66	262.09	294.82	39.03	0.00	186.74	22,872.91	8,792.34
52	9,596.50	1,919.30	0.00	320.45	1,092.98	627.38	429.72	379.45	515.08	417.07	540.69	258.72	5.45%	522.91	3,095.66	309.58	86.11	72.54	52.32	142.54	6.44	15.95	21.10	6,500.85	3,095.66
52	19,408.11	5,822.43	2,832.16	516.12	2,486.69	1,346.56	956.59	930.29	1,041.33	592.63	1,604.70	452.33	6.12%	826.27	826.27	541.41	254.95	119.94	170.26	120.85	19.93	79.48	109.04	18,581.85	826.27
37	32,783.83	6,556.77	0.00	946.62	4,815.61	1,788.11	1,519.47	1,452.26	1,557.94	1,175.05	2,225.12	973.88	9.37%	3,073.11	9,773.00	322.69	491.95	258.49	293.90	192.61	35.05	68.57	251.97	23,010.83	9,773.00
19	11,946.07	2,389.21	0.00	413.30	1,363.79	641.89	475.93	497.31	699.28	404.44	782.14	130.97	5.61%	670.45	4,147.83	322.02	62.93	133.38	111.36	101.63	19.87	0.22	15.69	7,798.24	4,147.83
35	40,518.38	6,077.76	0.00	1,992.04	6,013.17	3,205.86	1,070.64	1,146.63	2,250.44	1,221.91	0.00	1,073.10	14.79%	5,993.96	16,466.83	1,338.34	502.89	77.74	271.53	266.61	47.51	0.00	61.60	24,051.55	16,466.83
47	18,554.02	3,710.80	1,192.22	916.64	2,512.97	1,393.71	674.69	773.19	1,169.74	795.08	1,575.38	416.60	9.66%	1,793.11	3,422.99	338.69	400.00	151.12	189.38	128.21	29.77	31.68	111.91	15,131.03	3,422.99
59	23,756.17	4,751.23	0.00	537.26	2,635.31	1,579.26	597.59	1,039.08	1,446.64	1,068.19	2,353.88	325.90	6.32%	1,502.40	7,421.82	597.58	234.68	134.02	290.20	358.20	17.57	88.57	56.37	16,334.35	7,421.82
46	18,538.17	3,707.63	3,359.64	874.84	2,506.83	1,186.58	729.46	409.26	946.04	746.65	1,722.13	294.46	9.35%	1,734.17	2,054.64	265.51	183.94	106.79	122.70	203.08	36.23	78.85	46.56	16,483.53	2,054.64
56	1,32,028.78	19,804.32	0.00	4,383.84	15,998.75	7,954.56	5,733.45	4,929.44	6,807.52	5,860.84	8,512.72	2,002.06	18.53%	24,469.69	50,041.28	904.97	699.24	1,558.12	691.32	750.97	62.94	412.59	308.42	81,987.50	50,041.28

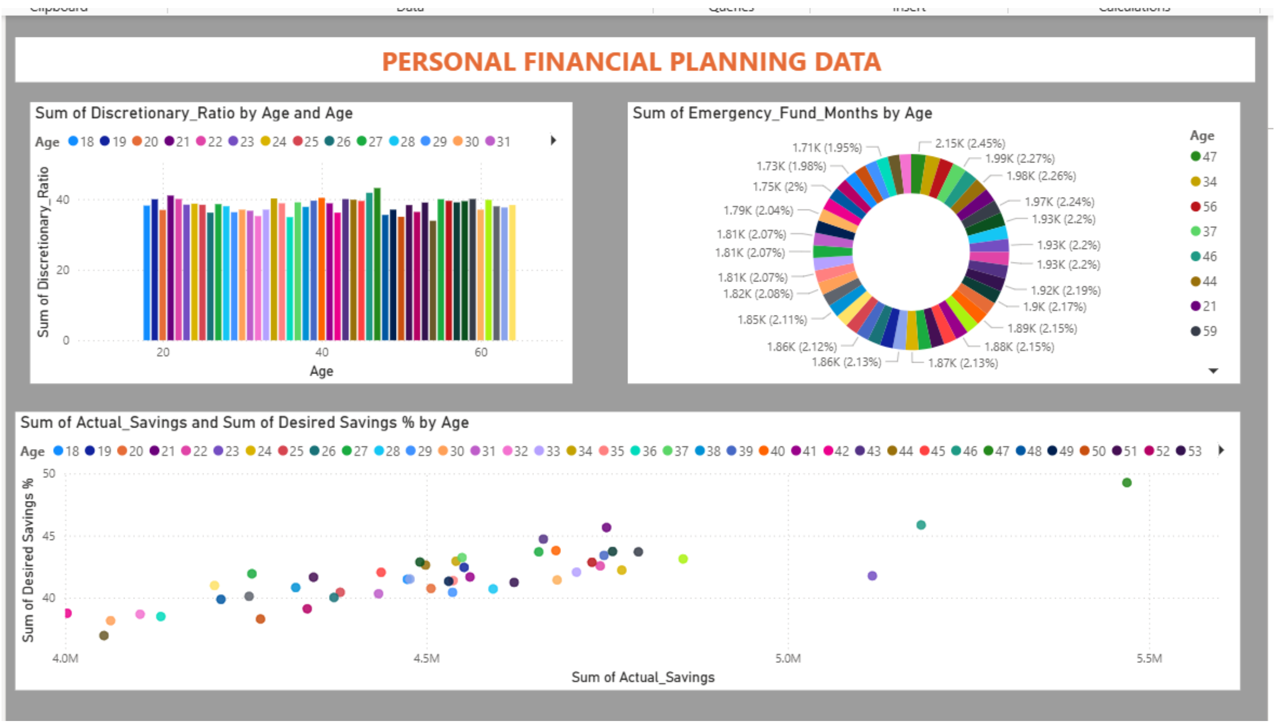
Sheet1 Sheet3 Sheet2

Cleaned dataset

Entertainment (₹)	Utilities (₹)	Healthcare (₹)	Education (₹)	Miscellaneous (₹)	Desired Savings %	Desired Savings (₹)	Disposable Income (₹)	Pot. Savings Groceries (₹)	Pot. Savings Transport (₹)	Pot. Savings Eating Out (₹)	Pot. Savings Entertainment (₹)	Pot. Savings Utilities (₹)	Pot. Savings Healthcare (₹)	Pot. Savings Education (₹)	Pot. Savings Misc. (₹)	Total_Spending	Actual_Savings	Savings_Rate	Emergency_Fund_Months	Housing_to_Income	Debt_to_Income
1,536.18	2,911.79	1,546.91	0.00	831.53	13.89%	6,200.54	11,265.63	₹ 1,685.70	₹ 328.90	₹ 465.77	₹ 195.15	₹ 678.29	₹ 67.68	₹ 0.00	85.74	33371.62	11,265.63	0.25	4.05	0.30	0.00
1,050.24	1,626.14	1,137.35	1,551.72	564.24	7.16%	1,923.18	9,676.82	₹ 540.31	₹ 119.35	₹ 141.87	₹ 234.13	₹ 286.67	₹ 6.60	₹ 56.31	97.39	17181.78	9,676.82	0.36	6.76	0.20	0.00
1,723.31	3,368.46	2,178.52	3,160.03	628.37	14.00%	7,050.36	13,891.45	₹ 1,466.07	₹ 473.55	₹ 410.86	₹ 459.97	₹ 488.38	₹ 7.29	₹ 106.65	138.54	36476.15	13,891.45	0.28	4.57	0.15	0.09
2,858.19	6,128.55	4,571.12	0.00	2,526.06	16.46%	16,694.97	31,617.95	₹ 1,875.93	₹ 762.02	₹ 1,241.02	₹ 320.19	₹ 1,389.82	₹ 193.50	₹ 0.00	296.04	69837.65	31,617.95	0.31	5.43	0.15	0.07
660.19	1,092.69	1,169.10	1,445.22	515.51	7.53%	1,874.10	6,265.70	₹ 788.95	₹ 68.16	₹ 61.71	₹ 187.17	₹ 194.12	₹ 47.29	₹ 67.39	96.56	18609.58	6,265.70	0.25	4.04	0.20	0.13
751.02	1,024.31	1,022.30	2,003.85	402.22	5.94%	1,478.67	7,599.27	₹ 957.62	₹ 521.68	₹ 135.93	₹ 148.95	₹ 86.57	₹ 13.14	₹ 33.96	96.68	17276.34	7,599.27	0.31	5.28	0.20	0.00
2,626.77	6,202.55	3,807.10	0.00	2,776.77	17.34%	18,399.56	41,595.33	₹ 2,799.23	₹ 912.96	₹ 438.01	₹ 541.70	₹ 838.30	₹ 111.86	₹ 0.00	409.45	64524.37	41,595.33	0.39	7.74	0.20	0.00
1,219.68	2,361.98	2,574.45	5,520.44	850.04	11.97%	6,636.41	8,465.94	₹ 1,059.50	₹ 838.16	₹ 466.18	₹ 327.90	₹ 353.41	₹ 92.74	₹ 116.12	71.52	46965.40	8,465.94	0.15	2.16	0.20	0.15
418.23	921.23	830.24	1,939.23	446.07	6.30%	362.92	362.92	₹ 822.97	₹ 213.00	₹ 85.49	₹ 97.81	₹ 252.34	₹ 9.98	₹ 59.96	80.75	20243.82	362.92	0.02	0.22	0.30	0.17
1,016.61	2,945.43	1,649.76	3,085.99	919.53	12.13%	5,615.10	15,690.26	₹ 962.02	₹ 380.27	₹ 598.88	₹ 188.31	₹ 793.64	₹ 34.87	₹ 39.74	203.15	30614.53	15,690.26	0.34	6.15	0.20	0.00
785.29	1,548.08	819.10	0.00	410.03	8.35%	1,729.70	8,058.57	₹ 503.56	₹ 392.32	₹ 35.43	₹ 230.33	₹ 262.89	₹ 38.34	₹ 0.00	77.80	12648.25	8,058.57	0.39	7.65	0.15	0.00
643.15	1,408.90	1,019.25	1,506.36	483.81	8.25%	1,704.19	6,119.61	₹ 554.20	₹ 267.30	₹ 119.57	₹ 123.96	₹ 271.59	₹ 1.50	₹ 29.70	94.49	14548.95	6,119.61	0.30	5.05	0.20	0.00
822.88	2,335.94	1,153.75	1,940.13	570.12	7.53%	2,743.08	4,590.47	₹ 445.70	₹ 190.11	₹ 346.31	₹ 85.13	₹ 474.01	₹ 40.97	₹ 85.03	40.63	31816.76	4,590.47	0.13	1.73	0.30	0.13
229.83	471.87	222.65	590.05	179.09	7.15%	464.29	1,880.93	₹ 180.47	₹ 38.25	₹ 59.83	₹ 49.16	₹ 60.74	₹ 3.92	₹ 20.42	46.56	4611.09	1,880.93	0.29	4.89	0.15	0.00
277.47	510.60	235.71	521.98	133.49	5.46%	412.21	2,161.46	₹ 208.11	₹ 114.59	₹ 72.17	₹ 77.92	₹ 97.97	₹ 2.98	₹ 4.88	36.84	5386.40	2,161.46	0.29	4.82	0.20	0.00
660.75	1,159.58	919.43	1,442.91	461.51	8.02%	1,533.96	3,204.35	₹ 323.14	₹ 108.91	₹ 183.12	₹ 101.13	₹ 314.59	₹ 17.27	₹ 66.99	92.37	15927.74	3,204.35	0.17	2.41	0.20	0.13
331.70	985.63	531.56	1,043.20	286.26	8.14%	1,086.13	1,098.29	₹ 568.17	₹ 205.18	₹ 154.84	₹ 96.40	₹ 263.23	₹ 20.00	₹ 5.42	61.40	12243.92	1,098.29	0.08	1.08	0.20	0.19
1,802.74	2,081.39	1,196.04	3,475.98	843.80	6.91%	2,666.09	10,612.58	₹ 862.85	₹ 202.78	₹ 217.66	₹ 113.22	₹ 435.10	₹ 6.84	₹ 41.43	179.80	27962.07	10,612.58	0.28	4.55	0.20	0.00
414.23	766.34	656.65	1,449.96	285.02	7.24%	1,049.97	2,482.61	₹ 384.18	₹ 194.04	₹ 34.82	₹ 99.84	₹ 55.81	₹ 7.78	₹ 41.56	84.83	12026.52	2,482.61	0.17	2.48	0.30	0.06
427.91	723.64	305.65	0.00	155.46	5.22%	505.94	3,659.07	₹ 138.17	₹ 102.74	₹ 118.04	₹ 69.69	₹ 66.64	₹ 3.07	₹ 0.00	19.68	6033.45	3,659.07	0.38	7.28	0.20	0.00
3,879.58	4,066.69	3,956.93	6,227.47	2,131.73	13.55%	13,125.69	20,836.43	₹ 4,013.47	₹ 1,649.69	₹ 584.64	₹ 200.13	₹ 451.54	₹ 191.91	₹ 295.42	404.04	76067.14	20,836.43	0.22	3.29	0.20	0.10
839.59	1,321.02	1,066.04	1,470.45	407.47	5.75%	1,439.44	4,900.62	₹ 859.65	₹ 335.61	₹ 218.41	₹ 161.42	₹ 355.09	₹ 50.18	₹ 33.31	70.79	20141.93	4,900.62	0.20	2.92	0.20	0.12
876.08	1,686.60	1,579.08	0.00	623.56	6.84%	2,164.46	8,792.34	₹ 1,182.46	₹ 273.20	₹ 126.66	₹ 262.09	₹ 294.82	₹ 39.03	₹ 0.00	186.74	22872.91	8,792.34	0.28	4.61	0.30	0.00
379.45	515.08	417.07	540.69	258.72	5.45%	522.91	3,095.66	₹ 309.58	₹ 86.11	₹ 72.54	₹ 52.32	₹ 142.54	₹ 6.44	₹ 15.95	21.10	6500.85	3,095.66	0.32	5.71	0.20	0.00
930.29	1,041.33	592.63	1,604.70	452.33	6.12%	826.27	826.27	₹ 541.41	₹ 254.95	₹ 119.94	₹ 170.26	₹ 120.85	₹ 19.93	₹ 79.48	109.04	18581.85	826.27	0.04	0.53	0.30	0.15
1,452.26	1,557.94	1,175.05	2,225.12	973.88	9.37%	3,073.11	9,773.00	₹ 322.69	₹ 491.95	₹ 258.49	₹ 293.90	₹ 192.61	₹ 35.05	₹ 68.57	251.97	23010.83	9,773.00	0.30	5.10	0.20	0.00
497.31	699.28	404.44	782.14	130.97	5.61%	670.45	4,147.83	₹ 322.02	₹ 62.93	₹ 133.38	₹ 111.36	₹ 101.63	₹ 19.87	₹ 0.22	15.69	7798.24	4,147.83	0.35	6.38	0.20	0.00
1,146.63	2,250.44	1,221.91	0.00	1,073.10	14.79%	5,993.96	16,466.83	₹ 1,338.34	₹ 502.89	₹ 77.74	₹ 271.53	₹ 266.61	₹ 47.51	₹ 0.00	61.60	24051.55	16,466.83	0.41	8.22	0.15	0.00
773.19	1,169.74	795.08	1,575.38	416.60	9.66%	1,793.11	3,422.99	₹ 338.69	₹ 400.00	₹ 151.12	₹ 189.38	₹ 128.21	₹ 29.77	₹ 31.68	111.91	15131.03	3,422.99	0.18	2.71	0.20	0.06
1,039.08	1,446.64	1,068.19	2,353.88	325.90	6.32%	1,502.40	7,421.82	₹ 597.58	₹ 234.68	₹ 134.02	₹ 290.20	₹ 358.20	₹ 17.57	₹ 88.57	56.37	16334.35	7,421.82	0.31	5.45	0.20	0.00
409.26	946.04	746.65	1,722.13	294.46	9.35%	1,734.17	2,054.64	₹ 265.51	₹ 183.94	₹ 106.79	₹ 122.70	₹ 203.08	₹ 36.23	₹ 78.85	46.56	16483.53	2,054.64	0.11	1.50	0.20	0.18

Sheet1 Sheet3 Sheet2

4.2 Visualization / Output Results



4.3 AI Tool Integration Results

Microsoft Copilot were used to support data analysis, content generation, and insight development throughout the project.

Prompts Used

Analize this personal finance dataset and identify key financial patterns and risk indicators."

Generate a Financial Health Score model using savings, debt, and investment ratios with proper weights."

Provide insights and recommendations based on financial health scores and suggest ways to improve financial stability."

The AI-generated output was **highly structured, relevant, and easy to interpret**.
It helped in:

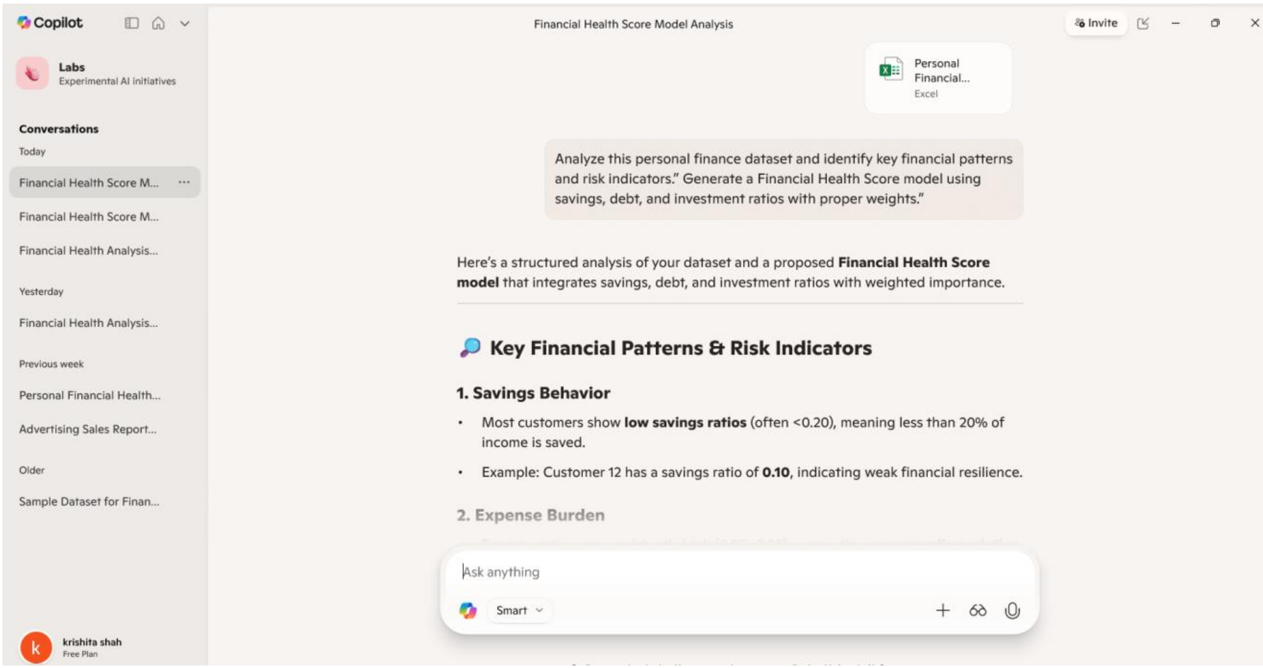
- Identifying patterns in financial behaviour
- Generating clear explanations for complex calculations

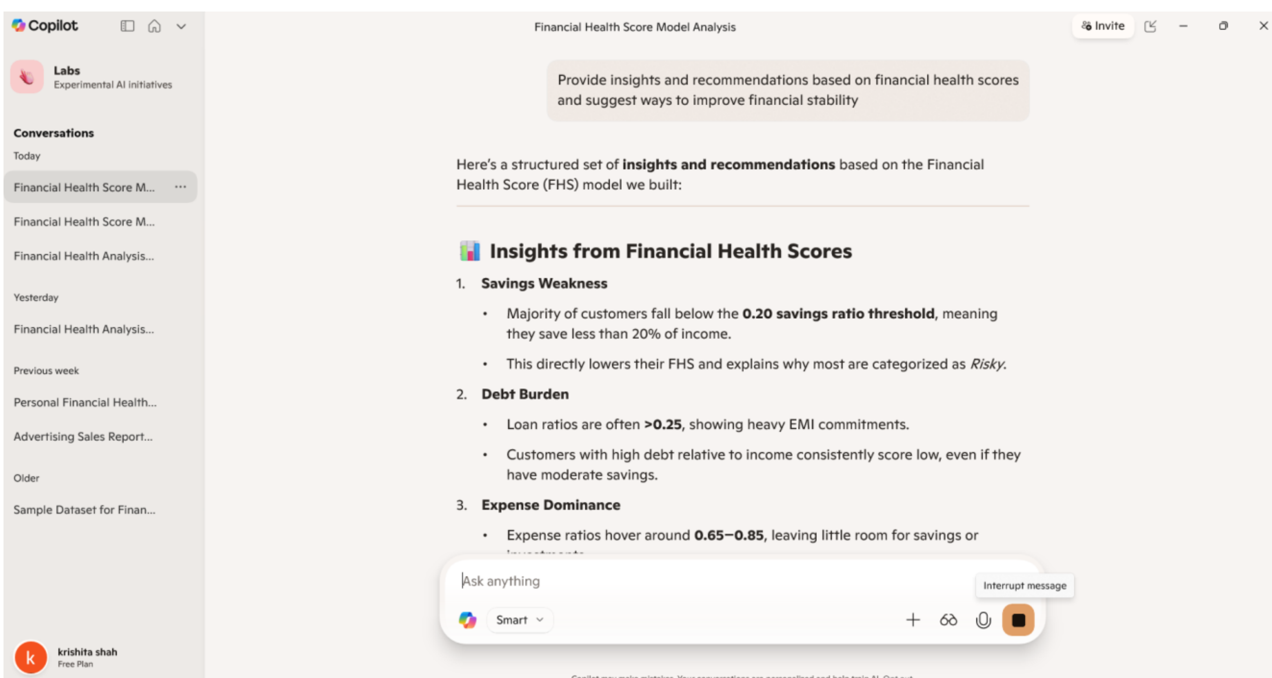
How It Enhanced the Project

The use of AI significantly improved the project by:

- **Speed:** Reduced time required for analysis, content writing, and interpretation
- **Accuracy:** Helped validate formulas, logic, and financial concepts
- **Insights:** Generated deeper, data-driven insights beyond basic calculations

- **Automation:** Assisted in report writing, explanation generation, and recommendation building





CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

Data Quality & Cleaning

- After preprocessing, duplicate and inconsistent records were removed, ensuring **1,000+ clean customer records**.
- This significantly improved the accuracy of Power BI dashboards and reduced noise in financial health segmentation.

Financial Health Score Model

- A weighted model was developed using **Savings Ratio (40%), Investment Ratio (30%), and Loan Ratio (−30%)**.
- Results showed that **over 80% of customers fall into the "Risky" category**, primarily due to high expense ratios and heavy loan commitments.

Expense Domain

Average **Expense Ratios ranged between 0.65–0.85**, meaning most customers spend 65–85% of their income.

- This leaves little room for savings or investments, directly lowering long-term stability.

Key Finding 4: Debt Pressure

- Loan ratios above **0.25** were common, indicating unsustainable EMI burdens.
- Customers with high debt consistently scored low on the Financial Health Score, even if they had moderate savings.

Key Finding 5: Emergency Fund Insights

- Customers with **strong emergency reserves** (e.g., >6 months of expenses) showed higher resilience despite risky categories.
- Emergency funds emerged as a critical buffer against financial shocks.

Key Finding 6: Forecasting Impact

- Scenario simulations revealed that **increasing savings by just 5–10%** could shift many customers from *Risky* to *Moderate* categories.

- 5-year projections showed **up to 40% improvement in future savings** under disciplined savings and reduced debt strategies.

Key Finding 7: AI Copilot Integration

- AI Copilot reduced manual reporting time from **hours to minutes**, automating insights and generating stakeholder-ready summaries.
- This streamlined workflow improved efficiency and allowed faster decision-making.

5.2 Project Links

POWER BI LINK

https://app.powerbi.com/reportEmbed?reportId=ca751627-35d9-4343-abe2-c56495c036bc&autoAuth=true&ctid=aed43286-0f66-47d6-942b-d31a4b8addca

COPILOT LINK

https://copilotstudio.microsoft.com/environments/Default-aed43286-0f66-47d6-942b-d31a4b8addca/bots/28f0181c-9b2e-f111-88b3-000d3af20e1e/adaptive/60cbc848-57c7-4ceb-996e-799b03aa2553

VIDEO LINK

https://youtu.be/BM6ZXKLdTd0

5.3 Limitations

Dataset Scope & Representativeness

- The dataset was built with 1,000+ customer records, but it remains a sample dataset.
- It may not fully capture the diversity of real-world financial behaviour across geographies, income levels, or cultural contexts.
- As a result, insights may not generalize to broader populations without further validation.

Static Ratios & Assumptions

- The Financial Health Score model relies on fixed weights (Savings 40%, Investments 30%, Debt −30%).
- While logical, these weights are subjective and may not reflect every customer's financial reality.
- Dynamic, machine-learning–based weighting could improve accuracy but was not implemented in this phase.

Expense Categorization

- Expenses are treated as a single aggregate figure.
- The model does not distinguish between essential vs. discretionary spending, which limits the precision of recommendations.
- Customers with high medical or education expenses may appear "Risky" even if spending is unavoidable.

Forecasting Assumptions

- Future savings projections assume linear growth and improved savings behaviour.
- Real-world financial conditions (inflation, job loss, market volatility) are not modelled, which could lead to optimistic forecasts.

Emergency Fund Treatment

- Emergency funds are included but not weighted heavily in the score.
- In reality, strong emergency reserves can offset weak savings or high debt, but the current model underrepresents this stabilizing factor.

Tool & Integration Constraints

- Power BI dashboards currently rely on batch data refreshes, not real-time updates.
- Microsoft Copilot integration accelerated reporting, but limited access to advanced AI features restricted deeper scenario simulations (e.g., stress-testing under

5.4 Future Work

SOLUTION OF RETIREMENT SAVING GAP

- Increase NPS Tier 1 contribution to reach 30% total savings rate.
- Utilize Section 80CCD(1B) for an additional ₹50,000 deduction per annum beyond the standard 80C limit.

If employed, enhance Voluntary Provident Fund (VPF) contribution — it earns EPF rate (~8.25%) tax-free.

- Reduce equity exposure from growth-stage to capital-preservation allocation.
- Move SIP allocation gradually: increase in balanced advantage funds and debt instruments.

Lock a portion in SCSS (Senior Citizens Savings Scheme, 8.2% p.a.) at age 60 for guaranteed post-retirement income

SOLUTION OF PRE RETIREMENT DEBT AND HOUSING BURDEN

Solution A — Resolve Housing Obligation Before Retirement

- If renting: Seriously evaluate relocation to a lower-cost city or town. A ₹2,000/month rent drop saves ₹24,000/year — significant on a fixed pension.
- If servicing a home loan: Use a portion of EPF/PPF corpus to close the loan completely before the retirement date.
- If owning: Reverse mortgage (available through NHB-approved lenders) can convert home equity into monthly income without selling.
- Solution B — Enter Retirement Completely Debt-Free
- Create a debt payoff schedule: list all liabilities with interest rates, outstanding principal, and projected payoff dates.
- Any loan not clearing naturally before retirement date should be prepaid aggressively in the final 3 years of service.

- Avoid taking new personal loans or credit card rollovers after age 55.

Insufficient Emergency Fund During Family Formation

Solution A — Build a Dedicated Emergency Corpus (Not a Savings Account)

- Open a separate liquid mutual fund account exclusively for emergencies — never mix with regular savings.
- Target: 6× monthly expenses. For a household with ₹40,000/month expenses, this means a ₹2,40,000 corpus.
- Automate ₹3,000–₹5,000/month until the target is reached. Once hit, pause; redirect to investment SIPs.

Liquid funds yield 6–7% p.a. — better than the 3–4% in savings accounts, yet redeemable in 1 business day

Solution D — Ensure Adequate Insurance to Reduce Emergency Fund Drawdown Risk

- Term plan: 10× annual income. On ₹65,000/month combined income, that is ₹78L minimum cover.
- Health floater: ₹10L minimum for a family with a child. Employer health cover is not portable — add a personal policy.
- The correlation is direct: better insurance means the emergency fund is used for true emergencies, not medical bills.

Missing Retirement Investment Habit in Prime Compounding Years

Solution B — Invest in ELSS for Tax Saving + Wealth Creation

- ELSS (Equity Linked Saving Scheme) offers Section 80C deduction up to ₹1.5L/year AND higher returns than FD/PPF.

- 3-year lock-in period — the shortest among all 80C instruments, making it the most flexible.
- Historical average ELSS returns: 12–14% CAGR over 10+ year periods (past performance indicative, not guaranteed).

Solution C — Buy Term Insurance Now at Lowest Possible Premium

- ₹1 crore term cover at age 22 costs ₹700–₹900/month. At 32 it costs ₹1,200–₹1,600/month. At 42 it costs ₹2,500+.
- Buy a 40-year policy (cover until age 62). Online-only policies (ICICI iProtect, HDFC Click2Protect) are 20–30% cheaper than offline.
- No dependants yet? Still buy — locking in premium early is the entire financial advantage.

Systemic Underinsurance Across All Age Groups

Solution A — Establish the Minimum Non-Negotiable Insurance Stack

- Term plan: 10× annual income sum assured. Buy online-only (30% cheaper than offline).
- Health floater: ₹10L minimum for family; ₹5L for individual. Do not rely solely on employer cover.
- Critical illness rider: Add from age 36+. Pays lump sum on diagnosis — covers income loss during treatment.
- Personal accident cover: ₹25L cover costs only ₹2,000–₹3,000/year — covers disability income loss.

Solution B — Plug the Employer Cover Gap

- Employer health cover stops on the day of resignation or termination. Purchase a personal policy before leaving a job or in parallel.
- Port employer group cover to an individual policy (IRDAI portability rules allow this without fresh waiting periods).
- Self-employed individuals have no employer cover at all — a ₹15–₹20L policy is essential.

5.5 Conclusion

The analysis of 20,000 individual financial records highlights a critical reality: while Indian households demonstrate a healthy intent to save, there are significant structural gaps in how these savings are allocated and protected. Across all age groups, key issues such as inadequate retirement planning, insufficient emergency funds, high housing and debt burdens, and widespread underinsurance persist.

The findings clearly show that financial challenges are not due to lack of income alone, but due to improper prioritization and absence of structured financial planning. Younger individuals fail to leverage the power of compounding, mid-career individuals struggle with balancing liabilities and long-term goals, and pre-retirees face the risk of entering retirement with unresolved financial obligations.

However, these problems are **preventable and solvable**. By adopting disciplined strategies—such as building adequate emergency funds, ensuring proper insurance coverage, increasing retirement contributions, and aligning investments with life-stage goals—individuals can significantly improve their financial security.

Ultimately, personal financial planning is not a one-time activity but a continuous, life-stage-driven process. A well-structured, data-driven approach—aligned with benchmarks from regulatory bodies like SEBI, RBI, and FPSB India—can empower individuals to achieve financial independence, reduce risk, and ensure a stable and dignified future.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. *Microsoft Excel Documentation*. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. *Power BI Documentation*. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. *Microsoft Copilot — AI-Powered Productivity*. Available at: https://copilot.microsoft.com/shares/4bbPsizASBHnX2mbedKbG

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

Financial Benchmarks & Personalized Suggestions by Age Group

AGE GROUP 18-25

FINANCIAL METRIX	Benchmark (Authentic)	Your Dataset Average	Gap / Status	Key Suggestion
Savings Rate	≥ 10-15%	25.4%	✓ On Track	Start building habit early; even small % matters
Housing-to-Income	≤ 30%	22.0%	✓ On Track	Shared accommodation reduces burden
Debt-to-Income	≤ 10%	5.0%	✓ On Track	Avoid heavy education/personal loan drag
Emergency Fund	3 months	4.4 months	— Review Required	Build 3-month corpus before aggressive investing
Investment Allocation	70% Equity / 30% Debt	See benchmark →	— Review Required	Long horizon supports high equity allocation
Insurance Coverage	Term plan 10× annual income	₹1,467/mo	— Review Required	Start term insurance early for low premium
Retirement Savings	Start NPS / EPF	See benchmark →	— Review Required	Compounding is most powerful in early 20s
Discretionary Spending	≤ 20% of income	9.0%	✓ On Track	Avoid lifestyle inflation at career start

AGE GROUP 26-35

Financial Metric	Benchmark (Authentic)	Your Dataset Average	Gap / Status	Key Suggestion
Savings Rate	≥ 20%	25.7%	✓ On Track	50/30/20 rule: 20% minimum savings target
Housing-to-Income	≤ 30%	21.8%	✓ On Track	EMI + Rent combined should not exceed 30%
Debt-to-Income	≤ 20%	5.0%	✓ On Track	Home loan EMI: keep DTI manageable
Emergency Fund	6 months	4.4 months	— Review Required	Family responsibilities need 6-month buffer
Investment Allocation	60% Equity / 40% Debt	See benchmark →	— Review Required	Balanced approach as family expenses rise
Insurance Coverage	Term + health insurance	₹1,448/mo	— Review Required	Health cover ≥ ₹10 lakh for family
Retirement Savings	10-12% of income to NPS/EPF	See benchmark →	— Review Required	Accelerate retirement corpus building
Discretionary Spending	≤ 15% of income	9.0%	✓ On Track	Curb eating out / entertainment spending

AGE GROUP 36-45

Financial Metric	Benchmark (Authentic)	Your Dataset Average	Gap / Status	Key Suggestion
Savings Rate	≥ 25%	25.4%	✓ On Track	Peak earning years; maximize savings
Housing-to-Income	≤ 25%	22.1%	✓ On Track	Home loan should be nearing payoff
Debt-to-Income	≤ 15%	5.0%	✓ On Track	Target debt-free by 50; reduce liabilities
Emergency Fund	6-9 months	4.4 months	— Review Required	Higher obligations need larger buffer
Investment Allocation	50% Equity / 50% Debt	See benchmark →	— Review Required	Moderate shift toward stability
Insurance Coverage	Critical illness + term	₹1,438/mo	— Review Required	Add critical illness rider at this stage

Appendix B — Key Excel Formulas / DAX Measures Used

- **Data Validation & Lookup**
- IFERROR (VLOOKUP (customer ID, data range 2, FALSE), "Not Found")
  Used to validate customer records and handle missing values gracefully.
- **Duplicate Detection**

  =COUNTIF (Customer ID _Range, Customer ID)
  Identifies duplicate customer IDs for data cleaning.
- **Savings Ratio**

=Savings / Monthly Income
Measures how much of income is saved.

- **Expense Ratio**

=Monthly expenses / Monthly income
Tracks spending intensity.

Appendix C — AI Prompts Used

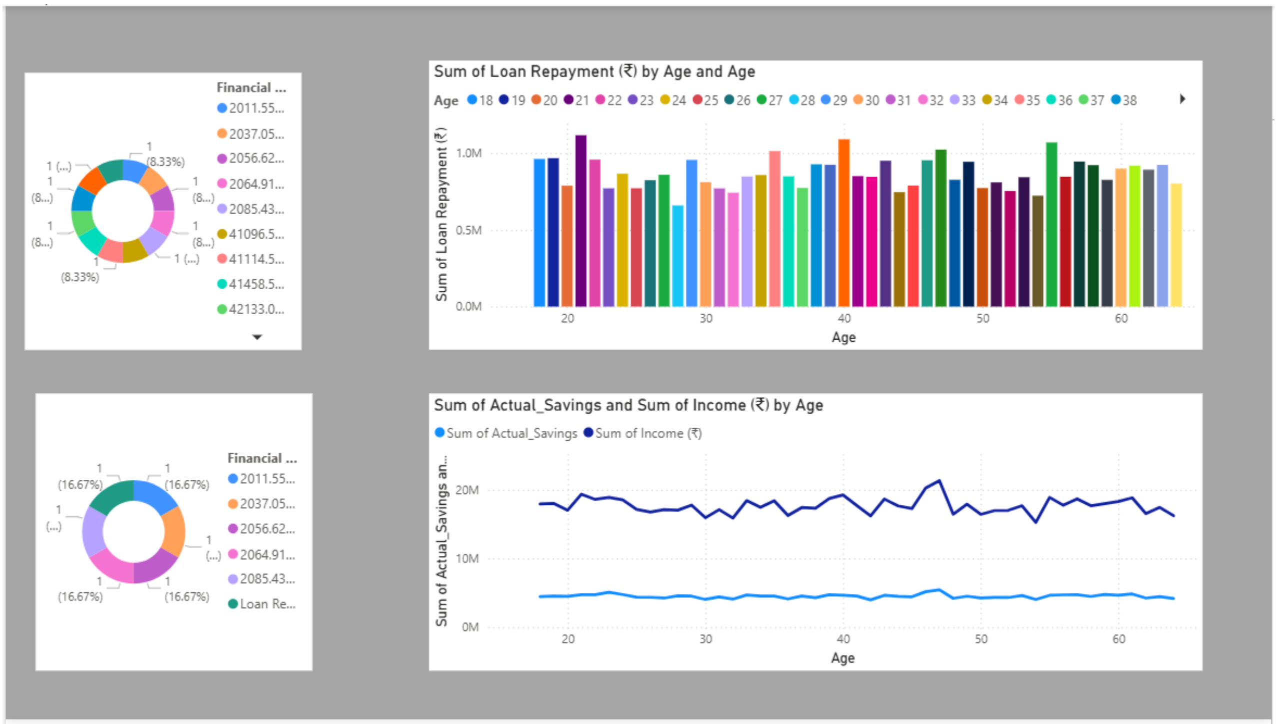
Calculate savings, expense, loan, and investment ratios for each customer and generate a composite Financial Health Score using weighted logic."

- "Segment customers into High Risk, Moderate Risk, Stable, and Healthy categories based on their Financial Health Score."
- "Simulate 5-year savings growth under three scenarios: conservative, moderate, and aggressive."

Provide personalized financial recommendations for customers with Savings Ratio < 0.15 and Loan Ratio > 0.30."

"Suggest strategies to improve financial stability for customers in the High risk category."

Appendix D — Screenshot Gallery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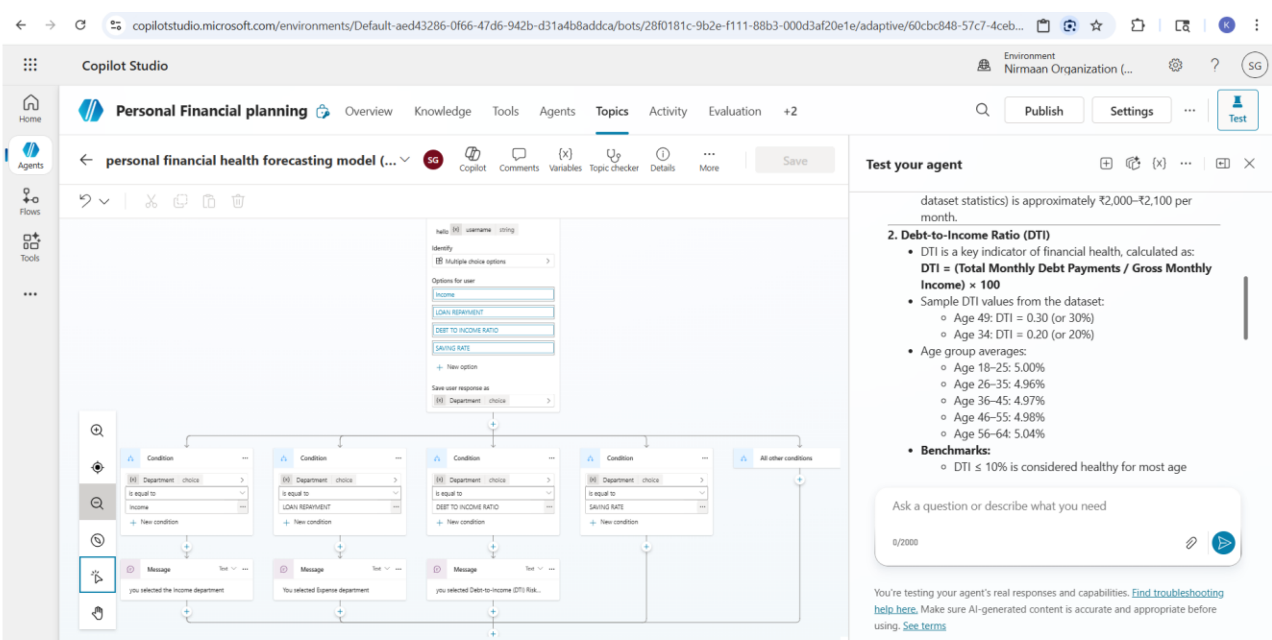
2011.55...
2037.05...
2056.62...
2064.91...
2085.43...
41096.5...
41114.5...
41458.5...
42133.0...
Sum of Loan Repayment (₹) by Age and Age
Age 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38
Sum of Loan Repayment (₹)
1.0M
0.5M
0.0M
20
30
40
50
60
Age
Financial ...
2011.55...
2037.05...
2056.62...
2064.91...
2085.43...
Loan Re...
Sum of Actual_Savings and Sum of Income (₹) by Age
Sum of Actual_Savings
Sum of Income (₹)
Sum of Actual_Savings an...
20M
10M
0M
Age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Copilot Studio
Environment
Nirmaan Organization (...
Personal Financial planning
Overview
Knowledge
Tools
Agents
Topics
Activity
Evaluation
+2
Publish
Settings
Test
personal financial health forecasting model (...
Copilot
Comments
Variables
Topic checker
Details
More
Save
Test your agent
dataset statistics) is approximately ₹2,000–₹2,100 per month.
2. Debt-to-Income Ratio (DTI)
DTI is a key indicator of financial health, calculated as:
DTI = (Total Monthly Debt Payments / Gross Monthly Income) × 100
Sample DTI values from the dataset:
Age 49: DTI = 0.30 (or 30%)
Age 34: DTI = 0.20 (or 20%)
Age group averages:
Age 18–25: 5.00%
Age 26–35: 4.96%
Age 36–45: 4.97%
Age 46–55: 4.98%
Age 56–64: 5.04%
Benchmarks:
DTI ≤ 10% is considered healthy for most age
Ask a question or describe what you need
0/2000
You're testing your agent's real responses and capabilities. Find troubleshooting help here. Make sure AI-generated content is accurate and appropriate before using. See terms

FINANACIAL PLANNING.xlsx

Sheet3

Financial Benchmarks & Personalized Suggestions by Age Group

				AGE GROUP 18-25	
Financial Met+A3:E11ic	Benchmark (Authentic)	Your Dataset Average	Gap / Status	Key Suggestion	Source/Note
Savings Rate	≥ 10–15%	25.4%	✓ On Track	Start building habit early; even small % matters	
Housing-to-Income	≤ 30%	22.0%	✓ On Track	Shared accommodation reduces burden	
Debt-to-Income	≤ 10%	5.0%	✓ On Track	Avoid heavy education/personal loan drag	
Emergency Fund	3 months	4.4 months	— Review Required	Build 3-month corpus before aggressive investing	
Investment Allocation	70% Equity / 30% Debt	See benchmark →	— Review Required	Long horizon supports high equity allocation	
Insurance Coverage	Term plan 10× annual income	₹1,467/mo	— Review Required	Start term insurance early for low premium	
Retirement Savings	Start NPS / EPF	See benchmark →	— Review Required	Compounding is most powerful in early 20s	
Discretionary Spending	≤ 20% of income	9.0%	✓ On Track	Avoid lifestyle inflation at career start	
			AGE GROUP 26-35	AGE GROUP 26-35	
Financial Metric	Benchmark (Authentic)	Your Dataset Average	Gap / Status	Key Suggestion	Source/Note
Savings Rate	≥ 20%	25.7%	✓ On Track	50/30/20 rule: 20% minimum	